

University of Minnesota
School of Physics and Astronomy

2026 Fall Mathematics 8302
Differential Topology
Assignment Solution

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Question 1

Let $O(n) = \{A \in M_n(\mathbb{R}) \mid A^T A = I\}$ be the orthogonal group, where A^T is the transpose of A . Consider the map

$$\varphi : M_n(\mathbb{R}) \rightarrow \text{Sym}(n), \quad A \mapsto A^T A, \quad (1)$$

where $\text{Sym}(n)$ is the set of symmetric $n \times n$ matrices (i.e., $B^T = B$).

1. Show that $\text{Sym}(n)$ is a manifold. Compute its dimension.
2. Compute the derivative of φ and show that φ is a submersion.
3. Prove that $O(n)$ is a submanifold of $M_n(\mathbb{R})$. What is the dimension of $O(n)$?
4. Prove that $O(n)$ is compact.

Answer

Definition 7.1 (Submersion, Ko Honda's notes). Let $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^n$ be open sets. A smooth map $f : U \rightarrow V$ is a **submersion** if $df(x)$ is surjective for all $x \in U$. More generally, a smooth map $f : M \rightarrow N$ between manifolds is a submersion if all of its local representatives $\psi_\beta \circ f \circ \phi_\alpha^{-1}$ are submersions.

Corollary 7.4 (Regular value theorem, Ko Honda's notes). Let $f : M \rightarrow N$ be a smooth map, where $\dim M = m$ and $\dim N = n$. A point $y \in N$ is called a **regular value** of f if $df(x)$ is surjective for every $x \in f^{-1}(y)$. If y is a regular value of f , then $f^{-1}(y)$ is a submanifold of M of dimension $m - n$.

1.

Consider the map $\psi : \mathbb{R}^{n(n+1)/2} \rightarrow \text{Sym}(n)$ defined by

$$\psi(x_1, \dots, x_{n(n+1)/2}) = \begin{pmatrix} x_1 & x_2 & \cdots & x_n \\ x_2 & x_{n+1} & \cdots & x_{2n-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_n & x_{2n-1} & \cdots & x_{n(n+1)/2} \end{pmatrix}. \quad (2)$$

This map is a bijection, and its inverse is given by

$$\psi^{-1} \left(\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix} \right) = (a_{11}, a_{12}, \dots, a_{1n}, a_{22}, a_{23}, \dots, a_{2n}, \dots, a_{nn}). \quad (3)$$

We equip $\text{Sym}(n)$ with the subspace topology from $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$. Since ψ is linear, it is continuous, and ψ^{-1} is the restriction of a linear projection $M_n(\mathbb{R}) \rightarrow \mathbb{R}^{n(n+1)/2}$, hence also continuous. So ψ^{-1} is a homeomorphism.

Hence $\{(\text{Sym}(n), \psi^{-1})\}$ is an atlas consisting of a single chart. The only transition map is $\psi^{-1} \circ \psi = \text{id}$, which is smooth.

Since $\text{Sym}(n)$ is a subspace of $\mathbb{R}^{n^2}$, which is Hausdorff and second countable, $\text{Sym}(n)$ is also Hausdorff and second countable.

Therefore, $\text{Sym}(n)$ is a manifold of dimension $n(n+1)/2$.

2.

Consider the limit

$$\lim_{t \rightarrow 0} \frac{\varphi(A + tB) - \varphi(A)}{t} \quad (4)$$

$$= \lim_{t \rightarrow 0} \frac{(A + tB)^T(A + tB) - A^T A}{t} \quad (5)$$

$$= \lim_{t \rightarrow 0} \frac{A^T A + t(B^T A + A^T B) + t^2 B^T B - A^T A}{t} = B^T A + A^T B. \quad (6)$$

It is easy to see that $B^T A + A^T B$ is symmetric. Then the derivative of φ at A is given by

$$d\varphi(A) : M_n(\mathbb{R}) \rightarrow \text{Sym}(n), \quad B \mapsto B^T A + A^T B. \quad (7)$$

We show that $d\varphi(A)$ is surjective for every $A \in O(n)$. Consider $A \in O(n)$, then $A^T A = I$. Let $C \in \text{Sym}(n)$ be arbitrary. We want to find $B \in M_n(\mathbb{R})$ such that $B^T A + A^T B = C$. Choose $B = \frac{1}{2}AC$, then we have

$$d\varphi(A)(B) = B^T A + A^T B = \frac{1}{2}(AC)^T A + \frac{1}{2}A^T(AC) = \frac{1}{2}C^T A^T A + \frac{1}{2}A^T AC = \frac{1}{2}C^T + \frac{1}{2}C = C. \quad (8)$$

Hence $d\varphi(A)$ is surjective for every $A \in O(n) = \varphi^{-1}(I)$, i.e., I is a regular value of φ .

To be more clear, φ is a submersion at $A \in O(n)$, but not a submersion at $0 \in M_n(\mathbb{R})$, since $d\varphi(0) = 0$.

3.

By definition, $O(n) = \varphi^{-1}(I)$. Since I is a regular value of φ , by Corollary 7.4, $O(n)$ is a submanifold of $M_n(\mathbb{R})$ of dimension

$$\dim O(n) = \dim M_n(\mathbb{R}) - \dim \text{Sym}(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}. \quad (9)$$

4.

To show that $O(n)$ is compact, we need to show that it is closed and bounded by Heine-Borel theorem. Since $O(n) \subset M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, it suffices to show that $O(n)$ is closed and bounded in $\mathbb{R}^{n^2}$.

Since $\{I\}$ is closed in $\text{Sym}(n)$ and φ is continuous, $O(n) = \varphi^{-1}(\{I\})$ is closed.

To show that $O(n)$ is bounded, we need to show that there exists $M > 0$ such that $\|A\| \leq M$ for all $A \in O(n)$. Consider the Frobenius norm $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\text{Tr}(A^T A)}$. Since $A \in O(n)$, we have $A^T A = I$, so $\|A\|_F = \sqrt{\text{Tr}(I)} = \sqrt{n}$. Therefore, $O(n)$ is bounded.

Hence, $O(n)$ is compact. □

Question 2

Let $F(x_1, \dots, x_n)$ be a homogeneous function of degree d in n real variables, i.e.,

$$F(tx_1, \dots, tx_n) = t^d \cdot F(x_1, \dots, x_n). \quad (10)$$

1. Prove Euler's identity: $\sum_{i=1}^n x_i \frac{\partial F}{\partial x_i} = d \cdot F$.
2. Prove that the set $F^{-1}(a)$, for $a \neq 0$, is a submanifold of $\mathbb{R}^n$.

Answer

1.

Consider the function $g(t) = F(tx_1, \dots, tx_n)$. Then $g(t) = t^d \cdot F(x_1, \dots, x_n)$. Taking the derivative with respect to t , we have

$$g'(t) = \sum_{i=1}^n \frac{\partial F}{\partial x_i}(tx_1, \dots, tx_n) \cdot x_i = dt^{d-1} \cdot F(x_1, \dots, x_n). \quad (11)$$

Setting $t = 1$, we get

$$\sum_{i=1}^n x_i \frac{\partial F}{\partial x_i}(x_1, \dots, x_n) = d \cdot F(x_1, \dots, x_n). \quad (12)$$

2.

Assume F is smooth and $d > 0$. To show that $F^{-1}(a)$ is a submanifold of $\mathbb{R}^n$, we show that a is a regular value of F , i.e., $dF(x)$ is surjective for every $x \in F^{-1}(a)$.

Let $x \in F^{-1}(a)$, so $F(x) = a$. For $v \in \mathbb{R}^n$,

$$dF(x)(v) = \lim_{t \rightarrow 0} \frac{F(x + tv) - F(x)}{t} = \sum_{i=1}^n \frac{\partial F}{\partial x_i}(x) \cdot v_i. \quad (13)$$

Let $w \in \mathbb{R}$ be arbitrary. We want to find $v \in \mathbb{R}^n$ such that $dF(x)(v) = w$. First, taking $v = x$, by Euler's identity we have

$$dF(x)(x) = \sum_{i=1}^n x_i \frac{\partial F}{\partial x_i}(x) = d \cdot F(x) = d \cdot a \neq 0. \quad (14)$$

Hence, for $v = \frac{w}{d \cdot a} x$, by linearity of $dF(x)$,

$$dF(x)(v) = \frac{w}{d \cdot a} dF(x)(x) = \frac{w}{d \cdot a} \cdot d \cdot a = w. \quad (15)$$

Hence $dF(x)$ is surjective for every $x \in F^{-1}(a)$, i.e., a is a regular value of F . By Corollary 7.4, $F^{-1}(a)$ is a submanifold of $\mathbb{R}^n$ of dimension $\dim \mathbb{R}^n - \dim \mathbb{R} = n - 1$. $\square$

Remark. The assumption $d \neq 0$ is needed. If $d = 0$, then $F(tx) = F(x)$ for all t ; letting $t \rightarrow 0$ and using the continuity of F gives $F(x) = F(0)$, so F is constant. Then $F^{-1}(a)$ is either empty or all of $\mathbb{R}^n$, which is not of dimension $n - 1$.